

RETAILSPHERE (PTY) LTD

Data Platform Programme

Dimensional Modelling Design

Step 7 Deliverable — Dimensional Model Design

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1. Purpose

This document defines the dimensional modelling design of all fact and dimension tables for the RetailSphere Gold layer data platform. These tables were derived from all functional requirements stated in FRD-001 covering all business domains within the organisation. The dimensional model follows the Kimball methodology — star schema design with conformed dimensions shared across multiple data marts to ensure consistency and enable cross-domain analysis.

2. Scope

This document covers the conceptual design of all fact and dimension tables, covering 9 dimension tables and 7 fact tables across 6 business domains. It includes star schema diagrams and full column detail — column names, data types, descriptions, and sample values for each table.

This document does not cover physical database design — DDL scripts, indexes, constraints, and partitioning strategies are addressed in PDD-001 (Step 9 — Physical Database Design).

3. Dimensional Modelling Approach

The dimensional modelling approach used is the Kimball star schema — one central fact table surrounded by one level of dimension tables.

Three fact table types were implemented across the seven fact tables:

- Transaction fact tables (FactSales, FactFinance).
- Periodic snapshot fact tables (FactInventorySnapshot)
- Accumulating snapshot fact tables (FactReplenishment, FactProcurement, FactContactHistory)

FactSupplierContract does not cleanly fit any of the three types above — see section 7.2 note.

Both conformed dimensions (shared across multiple data marts) and domain-specific dimensions were used. Conformed dimensions include DimDate, DimProduct, and DimCustomer.

Two Slowly Changing Dimension strategies were applied:

- SCD Type 2 (history preserved): DimProduct, DimCustomer, DimStore, DimEmployee, DimAccount.
- SCD Type 1 (overwrite, no history): DimCampaign.

4. Conformed Dimensions

Conformed dimensions are dimension tables that are shared across different data marts. RetailSphere's conformed dimensions include DimDate, DimProduct and DimCustomer.

4.1 DimDate

DimDate is a conformed dimension shared across all 7 fact tables in the RetailSphere data platform. It provides a comprehensive set of date attributes supporting calendar and fiscal period analysis. DimDate is pre-populated programmatically and does not require SCD Type 2 as dates do not change.

Table 4.1 DimDate Dimension Table

Column Name	Data Type	Description	Sample value
date_key (PK)	int	The surrogate key of data dimension. This is the primary key of the date dimension. It is unique and not null.	2024
date	char(10)	Full date in mm/dd/yyyy format	06/09/2024
ansi_date	char(10)	Full date in yyyy-mm-dd format.	2024-06-09
sql_date	datetime	Full date in SQL Server datetime data type	09/06/2024 0:00:00
day	int	The day of the month	9
day_of_the_week	int	The day of the week; 1 for Monday, 7 for Sunday	2
day_name	varchar(9)	The name of the day in a week	Tuesday
day_of_the_year	int	The number of elapsed days since January 1, aka Julian date	48
week_number	int	The number of elapsed weeks since January 1, weeks start on Monday	7
month	int	The number of the month	2
month_name	varchar(9)	The full name of the month in English	February
short_month_name	char(3)	The short name of the month in English	Feb
quater	char(2)	Three-month period in calendar year, Qn format	Q1
year	int	The calendar year of the date	2008
fiscal_week	int	The number of elapsed weeks since September 1.	27
fiscal_period	char(4)	A period consists of four or five weeks depending on the fiscal calendar. Usually, the pattern used in this company is 454, 544, or 445. in FPN format. A fiscal year always consists of 12 periods.	FP9
fiscal_quarter	char (3)	A fiscal quarter is three fiscal periods, from September 1, FQn format	FQ2
fiscal_year	char (6)	Fiscal year begins on September 1, in FYnnnn format	FY2008
week_day	tinyint	1 if it is Monday to Friday; 0 otherwise	0
rsa_holiday	tinyint	1 if SA public holiday 0 otherwise	0
month_end	tinyint	1 if it is the last day in the month; 0 otherwise	0
period_end	tinyint	1 if it is last day in a fiscal period; 0 otherwise	0
short_tradingday	tinyint	1 if on that day the offline stores' opening hours are shorter than normal opening hours; 0 otherwise. This is applicable only for offline stores	0

4.2 DimProduct

DimProduct is a conformed dimension shared across FactSales, FactInventorySnapshot, FactProcurement, and FactSupplierContract. It defines the full product hierarchy – type, category, and subcategory and includes cross-reference codes for all three source systems (SAP, Shopify, and Oracle MICROS POS) as resolved by ADR-001. SCD Type 2 is applied to preserve historical product pricing for accurate historical margin reporting.

Table 4.2 DimProduct Dimension Table

Column Name	Data Type	Description	Sample Value
product_key	int	The surrogate key of the product dimension. It is unique it is not null, and is the primary key of product dimension	100
product_code	varchar(20)	Natural key. follows this format RS-[CATEGORY]-[SEQUENCE] for example: RS-CLT-00001 (clothing item 1), RS-HMW-00001 (homeware item 1), and RS-ELC-00001 (electronics item 1) Assigned by RetailSphere. Owned by Sipho's team. Never changes. All three source systems map to it.	RS-CLT-00001
name	varchar(30)	Product name	Classic Cotton Tee
description	varchar(100)	Product description	Unisex cotton t-shirt available in 6 colours and 5 sizes.
product_type	varchar(20)	Level 1 of the product hierarchy, such as clothing, electronics, homeware	Clothing
product_category	varchar(30)	Level 2 of the product hierarchy. Groups products within a type. For clothing: Tops, Bottoms, Footwear, Accessories. For homeware: Bedding, Kitchen, Bathroom. For electronics: Audio, Computing, Gaming.	Tops
product_subcategory	varchar(30)	Level 3 of the product hierarchy. Most granular product grouping. For Tops: T-shirts, Shirts, Hoodies, Jackets. For Bottoms: Jeans, Shorts, Trousers, Leggings. Used by Zanele for shrinkage reporting by subcategory — FR-INV-003.	T-shirts
unit_price	money	Price for one Item	500.00
unit_cost	money	Allocated direct and indirect costs	320.00
status	varchar(15)	Upcoming if we have signed contract with the supplier, Active if it is still being offered to the customer, and Expired if it is no longer being offered.	Active
sap_stock_code	varchar(20)	Product code as stored in SAP ECC material master. Part of ADR-001 cross-reference mapping.	MAT-001-BLK-L
shopify_sku	varchar(50)	Product SKU as stored in Shopify Plus product catalogue. Part of ADR-001 cross-reference mapping.	RS-TEE-BLK-L
pos_barcode	varchar(20)	EAN/UPC barcode scanned at Oracle MICROS POS checkout. Primary matching key for ADR-001 identity resolution.	4501234567890
supplier_code	varchar(20)	Links product to supplier in DimSupplier. Used for procurement price variance calculation FR-PRO-003.	SUP-001
effective_start_date	date	Date from which this version of the product record is valid. Populated when record is created or updated.	2024-01-01
effective_end_date	date	Date on which this version expired. NULL means currently active record.	NULL (current) or 2026-06-07
is_current	BIT	1 if this is the current active version of the product record. 0 if expired.	1
source_system_code	int	The key of the source system where this record was coming from.	2
create_timestamp	datetime	When the record was created in DDS	2024-01-01 0:00:00
update_timestamp	datetime	When the record was last updated.	2024-01-01 00:00:00

4.3 DimCustomer

DimCustomer is a conformed dimension shared across FactSales, FactContactHistory, and FactFinance. It covers

RetailSphere's 2.1 million loyalty members as well as anonymous in-store customers. Identity resolution across six source systems is governed by ADR-002, with a tiered matching hierarchy applied in the Silver layer. SCD Type 2 is applied to track customer segment changes over time. POPIA compliance fields including opt_out_status and popia_consent_date are embedded directly in this dimension.

Table 4.3 DimCustomer Dimension Table

Column Name	Data Type	Description	Sample Value
customer_key	int	The surrogate key of the customer dimension. It is unique it is not null, and is the primary key of customer dimension	100
customer_id	varchar(50)	Natural key. follows this format RS-CUST-[SEQUENCE]. Assigned by RetailSphere. Owned by Priya's team. Never changes. All three source systems map to it.	RS-CUST-0000001
customer_loyalty_number	int	The loyalty member number for the 2.1 million loyal members. 0 for non-loyal members	1245
first_name	varchar(100)	Customer's first name	John
last_name	varchar(100)	Customer's last name	Smith
gender	char(1)	M for Male; F for Female	M
email_address	varchar(200)	Customer's e-mail address.	john.smith@gmail.com
date_of_birth	date	Customer's date of birth	18-04-1973
id_number	varchar(20)	Customer's ID Number	73041895709081
address1	varchar(50)	First line of address	20
address2	varchar(50)	Second line of address	Piet Retief St
address3	varchar(50)	Third line of address	Wellington
address4	varchar(50)	Fourth line of address	
city	varchar(200)	Village, town, or city of the customer's residence.	Wellington
province	varchar(150)	The province of the customer's residence	Western Cape
postal_code	varchar(50)	Postal code	7654
country	varchar(250)	Country	South Africa
phone_number	varchar(30)	Telephone number including area code	(+27) 74 701 8535
occupation	varchar(100)	Customer's job/profession TBC with Priya if we need these	DBA
household_income	varchar(20)	Rang of household income in ZAR - TBC with Priya if we need these	50-60k
date_registered	datetime	The date when the customer first purchased or became a loyalty member	01-05-2024
status	varchar(10)	Active is the customer is currently a loyalty member or made a purchase in the last 12 months	Active
effective_start_date	date	For SCD type 2, when the data in the row becomes effective; this defaults to 01-01-1900	01-01-1900
effective_end_date	date	For SCD type 2, when the data in the row is no longer valid; this defaults to 31-12-2999.	31-12-2999
is_current	BIT	For SCD type 2, 1 if this is the current record; 0 otherwise	1
salesforce_loyalty_id	varchar(50)	customer identification from Salesforce	TBC

shopify_customer_id	varchar(50)	customer identification from Shopify	TBC
pos_receipt	varchar(50)	customer identification from POS	TBC
mk_cloud_id	varchar(50)	customer identification from Marketing Cloud	TBC
meta_pixel_id	varchar(50)	customer identification from Meta	TBC
ga4_client_id	varchar(50)	customer identification from Google Analytics 4	TBC
opted_out_status	BIT	1 if customer has opted out of all marketing; 0 if still opted in	0
popia_consent_date	date	Date customer gave consent for data processing. Required for POPIA compliance.	2024-01-01
is_anonymous	BIT	1 if the customer is an anonymous in-store transaction with no identity signal available. 0 if identity is known.	0
customer_segment	varchar(50)	One of four segments from FR-SAL-003: Anonymous in-store, Online only, Loyalty only, Full cross-channel. SCD Type 2 tracks segment changes.	Full cross-channel
create_timestamp	datetime	When the record was created in DDS.	01-05-2024
update_timestamp	datetime	When the record was updated.	01-05-2024

5. SALES DOMAIN

The Sales domain covers all customer purchase transactions across RetailSphere’s three channels – In-store POS, online Shopify, and mobile. The primary fact table is FactSales, supported by DimStore, DimEmployee, and DimChannel as domain-specific dimensions. Conformed dimensions DimDate, DimProduct, and DimCustomer are shared with other data marts.

5.1 FactSales

FactSales is a transaction fact table capturing one row per product line item per transaction. It is the most granular sales record in the platform, enabling customer lifetime value, channel performance, margin analysis, and discount effectiveness reporting as required by FR-SAL-001 to FR-SAL-005.

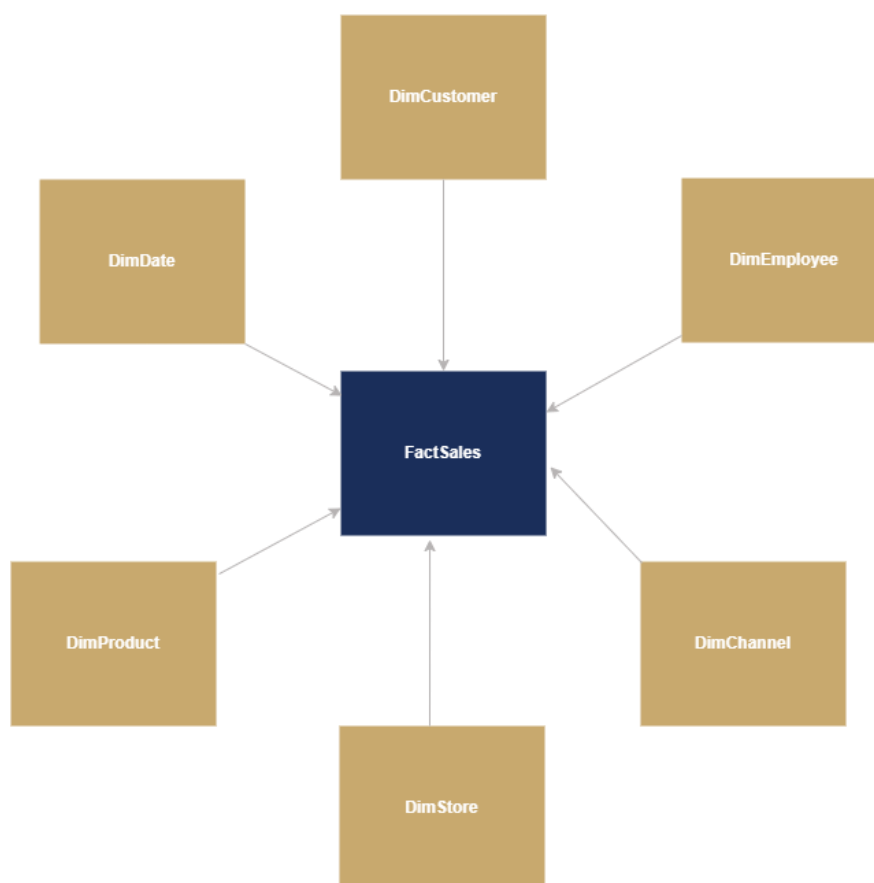


Figure 5.1 – FactSales Star Schema

Table 5.1 FactSales Fact Table

Column Name	Data Type	Description	Sample value
sales_date_key	int	FK to DimDate. Date the transaction occurred. Format YYYYMMDD.	20260601
customer_key	int	FK to DimCustomer. 0 for anonymous in-store transactions.	100
employee_key	int	FK to DimEmployee. Cashier who processed the transaction. 0 for online orders.	101
channel_key	int	FK to DimChannel. Sales channel — In-store POS or Online Shopify.	1
product_key	int	FK to DimProduct. Current version of product at time of sale.	200
store_key	int	FK to DimStore. Store where transaction occurred. 0 for online orders.	10
order_id	varchar(50)	Transaction or order identifier from source system.	POS-2026-001001
line_number	int	Line-item number within the transaction. Starts at 1.	1
quantity	int	Number of units sold on this line item.	2
unit_price	money	Selling price per unit at time of sale. Snapshot — not affected by future price changes.	550
unit_cost	money	Cost price per unit at time of sale. Used for margin calculation.	320
sales_value	money	Total line-item revenue. Calculated: quantity x unit_price.	1100
sales_cost	money	Total line-item cost. Calculated: quantity x unit_cost.	640
margin	money	Gross margin on this line item. Calculated: sales_value - sales_cost. Derived in dbt Gold model.	460

discount_amount	money	Rand value of discount applied to this line item. 0 if no discount.	110
discount_pct	decimal(5,2)	Percentage discount applied to this line item. 0.00 if no discount.	20
sales_timestamp	datetime	Exact date and time the transaction was processed at the till or online checkout.	2026-06-01 10:35:22
source_system_code	int	Key of the source system. 1 = Oracle MICROS POS. 2 = Shopify Plus.	1
create_timestamp	datetime	When the record was created in the data platform.	2026-06-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2026-06-01 0:00:00

5.2 DimStore

DimStore describes RetailSphere’s 87 stores nationally. It supports store-level performance analysis across sales, inventory, and finance domains. SCD Type 2 is applied to track store manager changes for historical performance attribution.

Table 5.2 DimStore Dimension Table

Column Name	Data Type	Description	Sample Value
store_key	int	Surrogate key of the store dimension. Unique, not null, primary key	10
store_id	varchar(20)	Natural key. follows this format RS-STR-[SEQUENCE]. Assigned by RetailSphere. Never changes	RS-STR-001
store_number	int	Internal store number used by operations team.	87
store_name	varchar(100)	Full trading name of the store	Canal Walk
store_type	varchar(50)	Type of store format. Full format, express, or outlet.	Full Format
region	varchar(50)	RetailSphere trading region. May differ from provincial boundary.	Cape Metro
store_address1	varchar(50)	First line of stored address.	Shop 214
store_address2	varchar(50)	Second line of stored address.	Canal Walk Shopping Centre
store_address3	varchar(50)	Third line of stored address.	Century City
store_address4	varchar(50)	Fourth line of stored address. Optional.	
city	varchar(100)	City where the store is located.	Cape Town
province	varchar(150)	Province where the store is located	Western Cape
postal_code	varchar(20)	Postal code of the store	7441
phone_number	varchar(30)	Store contact number including area code.	(+27) 21 555 1234
store_manager	varchar(100)	Full name of the current store manager SCD Type 2 tracks manager changes for performance analysis.	Thabo Nkosi
store_size_sqm	int	Floor size of the store in square meters. Used for stock capacity planning.	850
trading_hours	varchar(100)	Normal store operating hours. Links to DimDate short_trading_day column.	Mon-Sat 09h00-21h00, Sun 10h00-17h00
is_active	bit	1 if store is currently trading. 0 if closed or under renovations	1
effective_start_date	date	SCD Type 2 start date. Date from which this version of the store record is valid. Defaults to 01-01-1900	2020-03-01
effective_end_date	date	SCD Type 2 end date. NULL means currently active. Defaults to 31-12-2999	2999-12-31
is_current	bit	1 if this is the current active version of the store record. 0 if expired	1
source_system_code	int	Key of the source system where this record originated.	1

create_timestamp	datetime	When the record was created in the data platform.	2020-03-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2020-03-01 0:00:00

5.3 DimEmployee

DimEmployee describes all RetailSphere's employees who interact with the data platform – primarily in-store sales staff and cashiers. SCD Type 2 is applied to track store transfers and promotions. Salary data is excluded from this dimension and remains in the OLTP source system.

Table 5.3 DimEmployee Dimension Table

Column Name	Data Type	Description	Sample Value
employee_key	int	Surrogate key of the employee dimension. Unique, not null, primary key	101
employee_id	varchar(20)	Natural key. follows this format RS-EMP-[SEQUENCE]. Assigned by RetailSphere. Never changes	RS-EMP-101
first_name	varchar(100)	Employee's first name	Siphokazi
last_name	varchar(100)	Employee's last name	Booi
gender	char(1)	M for Male; F for Female	F
email_address	varchar(200)	Employee's email address	siphokazi.booi@retailshpere.co.za
phone_number	varchar(30)	Employee's contact number including are code	(+27) 72 704 1952
date_of_birth	date	Employee's date of birth	01-08-1997
id_number	varchar(20)	Employee's id number	9708015709080
address1	varchar(50)	First line of employee's address	11
address2	varchar(50)	Second line of employee's address	Thyme Str
address3	varchar(50)	Third line of employee's address	Bardale Village
address4	varchar(50)	Fourth line of employee's address	Kuilsriver
city	varchar(200)	Village, town, or city of the employee's residence.	Cape Town
province	varchar(150)	The province of the employee's residence	Western Cape
postal_code	varchar(50)	Postal code	7884
country	varchar(250)	Country	South Africa
employment_type	varchar(50)	Current Employee's title (in-store: full-time, casual/part-time)	full-time
title	varchar(100)	Current Employee's role (Store Manager, Cashier, Sales Rep, Team Leader)	Cashier
manager_id	varchar(50)	RS-EMP-[SEQUENCE] - Manager's employee number	RS-EMP-005
store_id	varchar(20)	Store ID where the employee works. Links to the DimStore dimension.	RS-STR-001
department_name	varchar(150)	The department where the employee belongs	Sales
source_system_code	int	Key of the source system where this record originated.	2
effective_start_date	date	SCD Type 2 start date. Date from which this version of the employee record is valid. Defaults to 01-01-1900	2020-03-01
effective_end_date	date	SCD Type 2 end date. NULL means currently active. Defaults to 31-12-2999	2999-12-31
is_current	bit	1 if this is the current active version of the employee record. 0 if expired	1
create_timestamp	datetime	When the record was created in the data platform.	3/1/2020 0:00:00

update_timestamp	datetime	When the record was last updated in the data platform.	3/1/2020 0:00:00
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5.4 DimChannel

DimChannel is a small reference dimension describing RetailSphere’s sales and marketing channels. It is used in both the Sales and CRM and Marketing domains.

Table 5.4 DimChannel Dimension Table

Column Name	Data Type	Description	Sample Value
channel_key	int	Surrogate key of the channel dimension. Unique, not null, primary key.	1
channel_code	varchar(20)	Natural key. Format RS-CHN-[SEQUENCE]. Assigned by RetailSphere. Never changes.	RS-CHN-001
channel_name	varchar(50)	Full name of the sales or marketing channel.	In-store POS
channel_type	varchar(50)	Category of channel. Sales: In-store, Online, Mobile. Marketing: Email, Social, Display, SMS. Attribution channels must align with ADR-006.	Sales
create_timestamp	datetime	When the record was created in the data platform.	2026-01-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2026-01-01 0:00:00

6. Inventory Domain

The inventory domain tracks the stock levels and replenishment across RetailSphere’s 87 stores. Two fact tables serve this domain – FactInventorySnapshot for daily stock level tracking and FactReplenishment for end-to-end replenishment pipeline monitoring. DimSupplier is introduced here as it primarily supports inventory and procurement domains.

6.1 FACTINVENTORYSNAPSHOT

FactInventorySnapshot is a periodic snapshot fact table capturing one row per product per store per day. It enables daily stock level monitoring, shrinkage calculation, and inventory value reporting as required by FR-INV-001 to FR-INV-005. Quantity measures are semi-additive and must be averaged – not summed across date ranges in reporting.

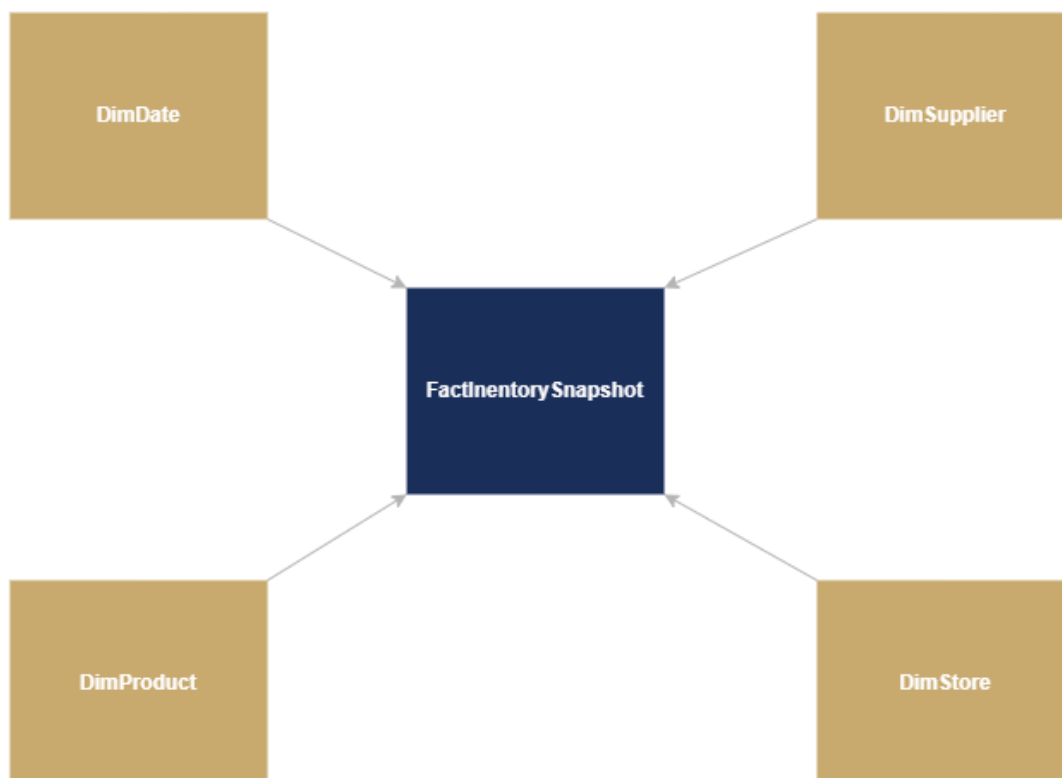


Figure 6.1 - FactInventorySnapshot Start Schema

Table 6.1 FactInventorySnapshot Fact Table

Column Name	Data Type	Description	Sample value
snapshot_date_key	int	FK to DimDate. Date the snapshot was taken. Format YYYYMMDD.	20260601
product_key	int	FK to DimProduct. Product being tracked.	200
store_key	int	FK to DimStore. Store holding this stock.	10
supplier_key	int	FK to DimSupplier. Primary supplier of this product to this store.	50
quantity_on_hand	int	Current stock quantity physically in the store at snapshot time. Semi-additive — average across dates.	48
quantity_on_order	int	Stock ordered from supplier but not yet received at store. Semi-additive.	100
quantity_reserved	int	Stock allocated to online orders not yet dispatched. Semi-additive.	5
unit_cost	money	Cost price per unit at time of snapshot. Used for inventory value calculation.	320
total_inventory_value	money	Total value of stock on hand. Calculated: quantity_on_hand x unit_cost. Derived in dbt Gold model.	15,360
snapshot_timestamp	datetime	Exact date and time the snapshot was captured from SAP and POS systems.	2026-06-01 23:00:00
source_system_code	int	Key of source system. 1 = SAP ECC. 2 = Oracle MICROS POS.	1
create_timestamp	datetime	When the record was created in the data platform.	2026-06-02 0:00:00

update_timestamp	datetime	When the record was last updated in the data platform.	2026-06-02 0:00:00
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6.2 FACTREPLENISHMENT

FactReplenishment is an accumulating snapshot fact table capturing one row per replenishment order per store per product. It tracks the full replenishment pipeline from reorder through DC pick, dispatch, and store receipt – enabling pipeline efficiency and supplier lead time analysis.

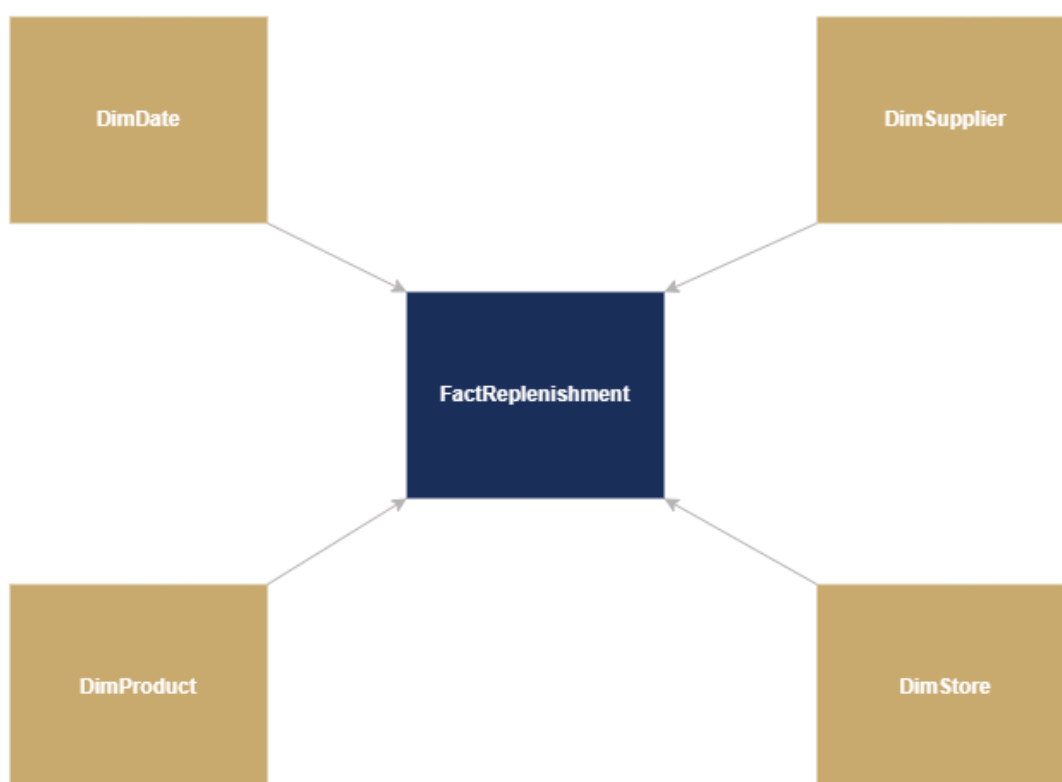


Figure 6.2 – FactReplenishment Star Schema

Table 6.2 FactReplenishment Fact Table

Column Name	Data Type	Description	Sample value
reorder_date_key	int	FK to DimDate. Date the snapshot was taken. Format YYYYMMDD.	20260601
product_key	int	FK to DimProduct. Product being replenished.	200
store_key	int	FK to DimStore. Store that needs stock replenishment.	50
supplier_key	int	FK to DimSupplier. Supplier fulfilling the replenishment order.	50
order_id	varchar(50)	Replenishment order identifier from SAP. Format RS-REP-[SEQUENCE].	RS-REP-001001
dc_pick_date	date	Date the Distribution Centre picked the stock. NULL if not yet picked.	2026-06-03

dispatch_date	date	Date stock left the DC and went in transit to store. NULL if not yet dispatched.	2026-06-04
store_receipt_date	date	Date stock arrived at store and GRN was captured in SAP. NULL if not yet received.	2026-06-07
days_to_pick	int	Days from reorder to DC pick. Calculated: dc_pick_date minus reorder_date. NULL until DC pick complete.	2
days_to_dispatch	int	Days from reorder to dispatch. Calculated: dispatch_date minus reorder_date. NULL until dispatch complete.	3
days_to_receipt	int	Days from reorder to store receipt. Calculated: store_receipt_date minus reorder_date. NULL until received.	6
total_days_pipeline	int	Total days from reorder to store receipt. Calculated: store_receipt_date minus reorder_date. NULL until complete.	6
quantity	int	Quantity of units ordered for replenishment.	200
order_status	varchar(20)	Current pipeline stage. Pending, Picked, In Transit, Received, Cancelled. Derived from populated date columns.	Received
source_system_code	int	Key of source system. SAP ECC replenishment orders.	1
create_timestamp	datetime	When the record was created in the data platform.	2026-06-01 0:00:00
update_timestamp	datetime	When the record was last updated. Updates at each pipeline stage completion.	2026-06-08 0:00:00

6.3 DIMSUPPLIER

DimSupplier describes RetailSphere’s 340 active suppliers. It is used across the Inventory, Procurement, and Finance domains. No SCD Type 2 is applied – supplier profile changes are overwritten using SCD Type 1.

Table 6.3 DimSupplier Dimension Table

Column Name	Data Type	Description	Sample Value
supplier_key	int	Surrogate key of the supplier dimension. Unique, not null, primary key.	50
supplier_code	varchar(20)	Natural key. Format RS-SUP-[SEQUENCE]. Assigned by RetailSphere. Never changes.	RS-SUP-001
supplier_name	varchar(100)	Full registered trading name of the supplier.	Cape Cotton Manufacturers (Pty) Ltd
email_address	varchar(200)	Primary supplier contact email address.	orders@capecotton.co.za
phone_number	varchar(30)	Supplier contact number including area code.	(+27) 21 555 7890
address1	varchar(50)	First line of supplier address.	Unit 5
address2	varchar(50)	Second line of supplier address.	Montague Gardens Industrial Park
address3	varchar(50)	Third line of supplier address.	Montague Drive
address4	varchar(50)	Fourth line of supplier address. Optional	
city	varchar(100)	City where the supplier is located	Cape Town
province	varchar(150)	Province where the supplier is located.	Western Cape
postal_code	varchar(20)	Postal code of the supplier.	7441
region	varchar(50)	RetailSphere trading region of the supplier.	Cape Metro
country	varchar(250)	Country where the supplier is registered.	South Africa
contact_person	varchar(150)	Primary account manager or contact at the supplier.	Thandi Eksteen
status	varchar(20)	Current supplier relationship status. Active — trading normally. Suspended — temporarily on hold. Blacklisted — no longer used.	Active

supplier_category	varchar(50)	Category of goods supplied. Clothing, Homeware, Electronics, Logistics, Packaging.	Clothing
create_timestamp	datetime	When the record was created in the data platform.	2020-01-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2024-06-01 0:00:00

7. Procurement Domain

The Procurement domain tracks RetailSphere’s supplier purchasing process from purchase order through to invoice payment. Two fact tables serve this domain – FactProcurement for end-to-end PO lifecycle tracking and FactSupplierContract for contracted rates and terms extracted from 340 active supplier contracts via ADR-003.

7.1 FACTPROCUREMENT

FactProcurement is an accumulating snapshot fact table capturing one row per purchase order line item per supplier. It tracks the full procurement pipeline from PO raised through GRN receipt, invoice matching, and payment – enabling supplier on-time delivery, invoice accuracy, price variance, and days payable outstanding analysis as required by FR-PRO-001 to FR-PRO-005.

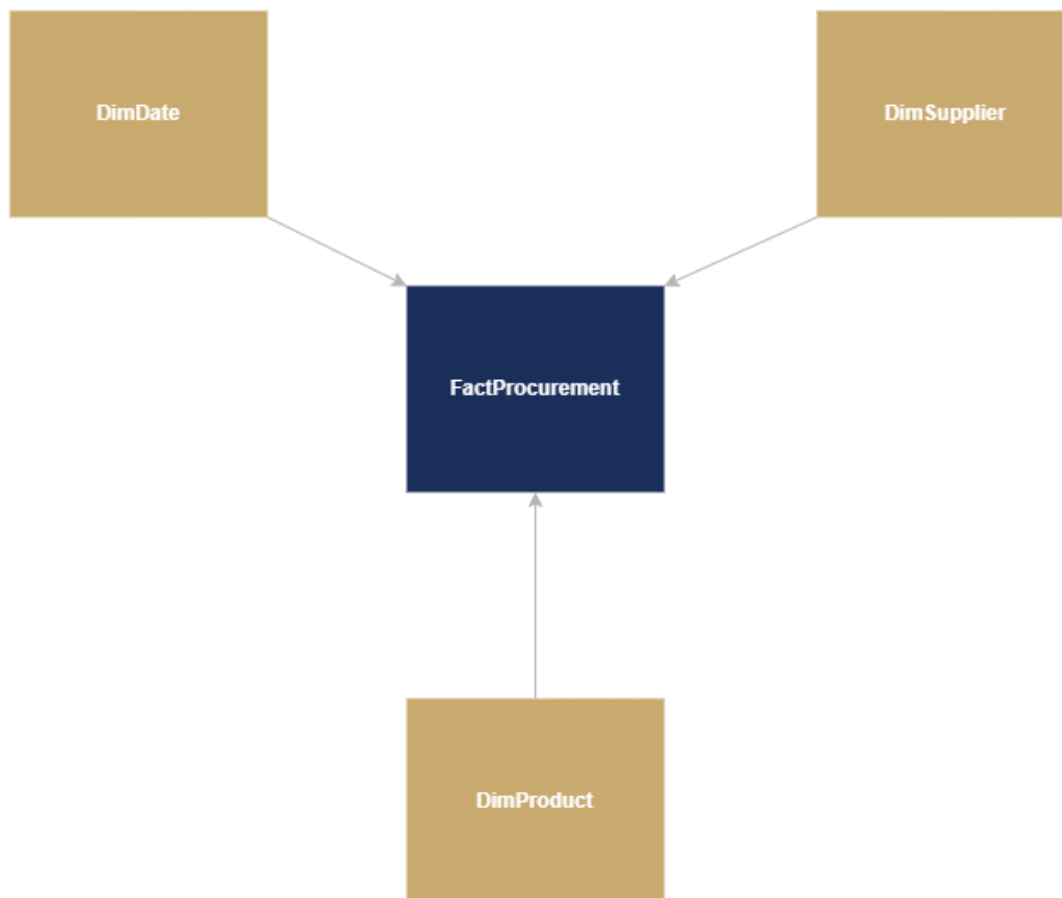


Figure 7.1 – FactProcurement Star Schema Diagram

Table 7.1 – FactProcurement Fact Table

Column Name	Data Type	Description	Sample value
po_date_key	int	FK to DimDate. Date the purchase order was raised. Format YYYYMMDD.	20260601
product_key	int	FK to DimProduct. Product being procured.	200
supplier_key	int	FK to DimSupplier. Supplier fulfilling the purchase order.	50
purchase_order_id	varchar(50)	Purchase order identifier from SAP. Format RS-PO-[SEQUENCE].	RS-PO-001001
grn_received_date	date	Date goods were received at DC or store. GRN captured in SAP. NULL if not yet received.	2026-06-05
invoice_received_date	date	Date supplier invoice was received. NULL if not yet received.	2026-06-06
invoice_matched_date	date	Date invoice was matched against PO and GRN (3-way match). NULL if not yet matched.	2026-06-07
invoice_paid_date	date	Date invoice was paid to supplier. NULL if not yet paid.	2026-06-14
delivery_date	date	Contracted delivery date from supplier contract. Used for on-time delivery calculation FR-PRO-001.	2026-06-05
days_to_delivery	int	Days from PO date to GRN received. NULL until GRN complete.	4
days_to_invoice	int	Days from GRN to invoice received. NULL until invoice received.	1
days_to_match	int	Days from invoice received to 3-way match complete. NULL until matched.	1
days_to_payment	int	Days from invoice matched to payment made. NULL until paid.	7
quantity	int	Quantity of units ordered on this PO line item.	500
unit_cost	money	Actual cost paid per unit. Sourced from SAP invoice.	330
contracted_rate	money	Contracted price per unit from supplier contract PDF. ADR-003 dependency.	320
price_variance	money	Variance between actual and contracted rate. Calculated: unit_cost minus contracted_rate. Derived in dbt Gold.	10
price_variance_pct	decimal(5,2)	Price variance as percentage of contracted rate. Calculated: price_variance / contracted_rate x 100.	3.13
total_order_value	money	Total value of this PO line item. Calculated: quantity x unit_cost.	165,000
total_days_pipeline	int	Total days from PO raised to invoice paid. NULL until payment complete.	13
po_status	varchar(20)	Current procurement stage. Ordered, Delivered, Invoiced, Matched, Paid, Disputed. Derived from populated date columns.	Paid
source_system_code	int	Key of source system. SAP ECC procurement records.	1
create_timestamp	datetime	When the record was created in the data platform.	2026-06-01 0:00:00
update_timestamp	datetime	When the record was last updated. Updates at each procurement stage completion.	2026-06-15 0:00:00

7.2 FACTSUPPLIERCONTRACT

FactSupplierContract resolves the many-to-many relationship between suppliers and products. One row per supplier per product per contract captures contracted rates, lead times, and payment terms extracted from PDF contracts via ADR-003. This table is the source of `contracted_rate` used in FR-PRO-003 price variance calculation in FactProcurement.

Note on fact table type: FactSupplierContract does not fit any of the three fact table types defined in section 3. Rows are immutable at creation except for **contract_end_date**, which transitions from NULL to a populated date only on contract row (new **contract_id**) rather than an update to an existing row's **product_key**, preserving historical rate integrity for FR-PRO-003 price variance calculations in FactProcurement. Treated in PDD-001 as an atypical transaction fact with a single permitted state transition.

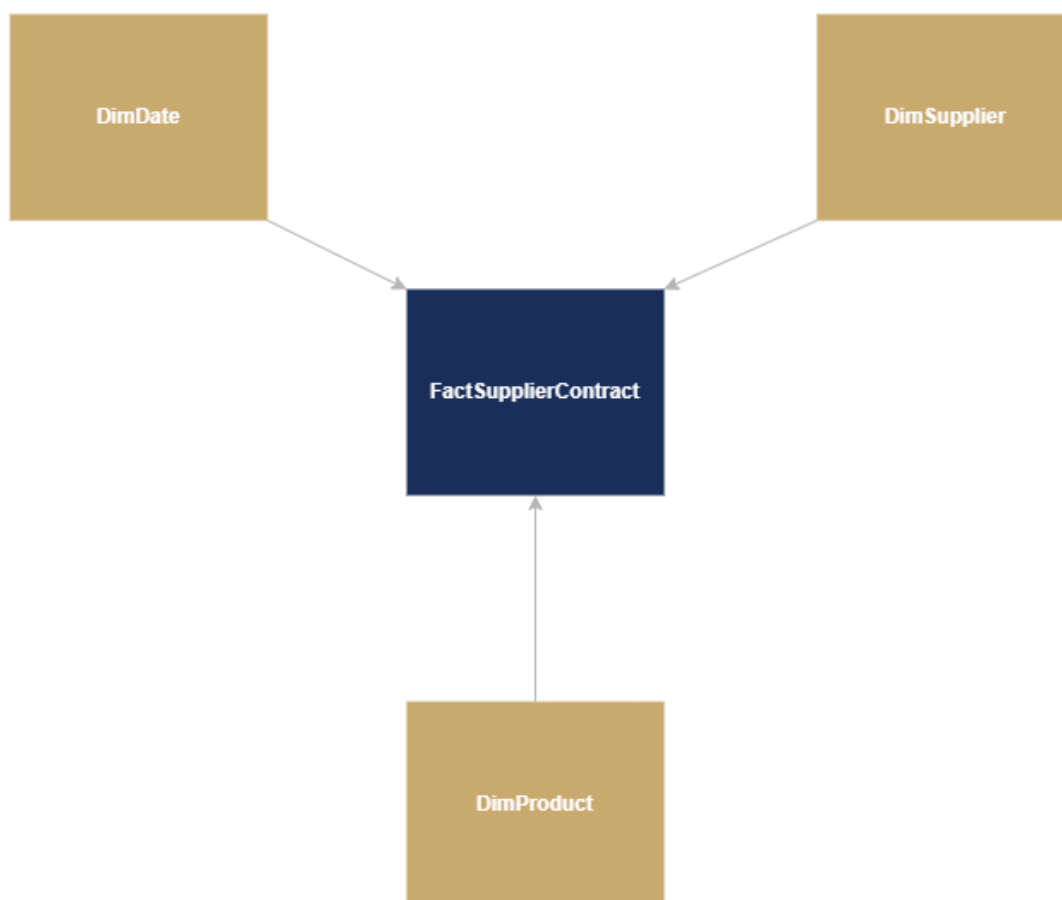


Figure 7.2 – FactSupplierContract Start Schema Diagram

Table 7.2 FactSupplierContract Fact Table

Column Name	Data Type	Description	Sample value
contract_start_date_key	int	FK to DimDate. Date this contract became effective. Format YYYYMMDD. Primary date dimension link.	20260601
supplier_key	int	FK to DimSupplier. Supplier party to this contract.	50
product_key	int	FK to DimProduct. Product covered by this contract.	200

contract_id	varchar(50)	Unique contract identifier. Format RS-CON-[SEQUENCE]. Extracted from supplier contract PDF via ADR-003.	RS-CON-001001
contract_end_date	date	Date this contract expires. NULL if open-ended contract.	2026-12-31
quantity	int	Minimum order quantity agreed in this contract.	500
contracted_rate	money	Agreed price per unit for this product under this contract. Extracted from PDF via ADR-003. Used in FR-PRO-003 price variance calculation.	320
lead_time	int	Agreed number of days from purchase order to delivery. Extracted from PDF via ADR-003.	7
payment_terms	varchar(20)	Agreed payment terms. 30 days, 60 days, 90 days. Extracted from PDF via ADR-003.	30 days
source_system_code	int	Key of source system. ADR-003 PDF extraction pipeline.	9
create_timestamp	datetime	When the record was created in the data platform.	2026-01-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2026-01-01 0:00:00

8. CRM AND MARKETING

The CRM and Marketing domain tracks all customer communications and campaign activity across RetailSphere’s marketing channels. FactContactHistory serves both the CRM and Marketing domains, capturing every contact attempt per customer per campaign per channel. This table enforces POPIA compliance through the 4-contact monthly cap and opt-out suppression logic governed by ADR-007.

8.1 FACTCONTACTHISTORY

FactContactHistory is an accumulating snapshot fact table capturing one row per customer per campaign per channel per contact attempt. It tracks the full communication lifecycle from sent through to delivery, opened, clicked, bounced, and unsubscribed. Suppressed contact attempts are also recorded with a suppression reason – enabling Priya’s team to audit POPIA compliance and Rethabile’s team to measure campaign attribution as required by FR-CUS-001, FR-CUS-002, and ADR-006.

No attempt-sequencing column exists to distinguish multiple contact attempts to the same customer/campaign/channel on the same day. PDD-001 v1.0 proceeds on the assumption of at most one attempt per day per combination. This assumption requires validation against actual Marketing Cloud send behaviour before build; unresolved as of this revision.

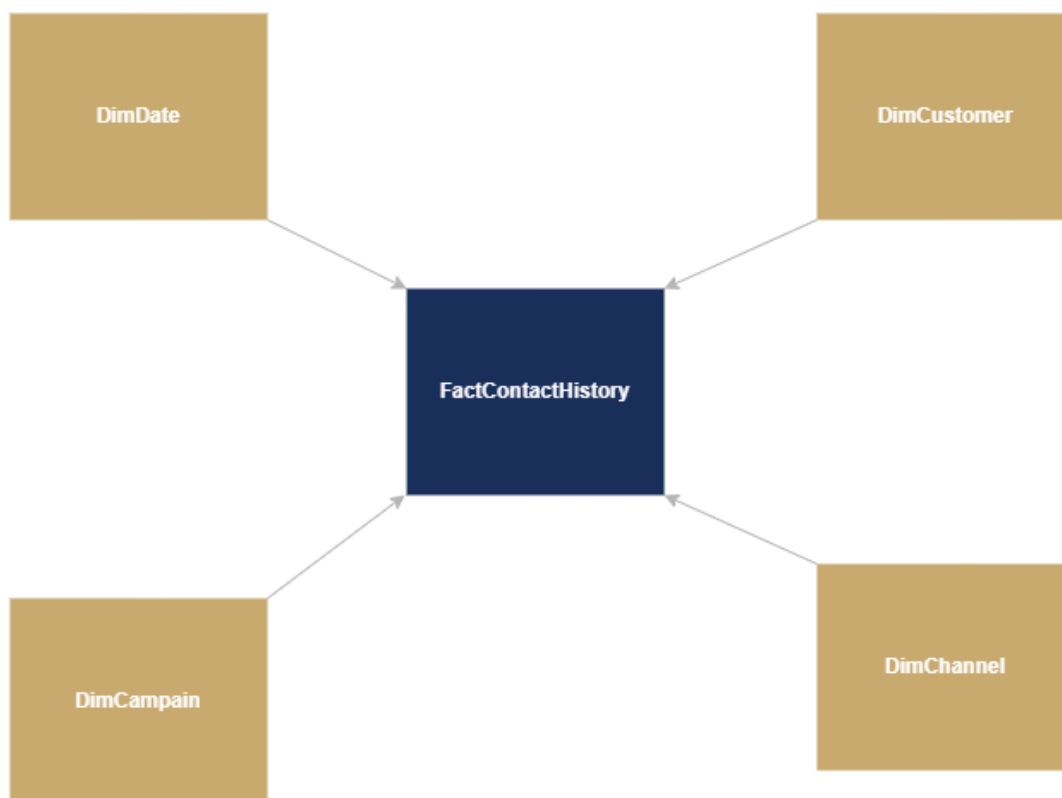


Figure 8.1 – FactContactHistory Star Schema

Table 8.1 FactContactHistory

Column Name	Data Type	Description	Sample value
contact_attempt_date_key	int	FK to DimDate. Date this contact attempt occurred or was attempted. Format YYYYMMDD.	20260601
customer_key	int	FK to DimCustomer. Member who was contacted.	100
campaign_key	int	FK to DimCampaign. Campaign that triggered this contact.	200
channel_key	int	FK to DimChannel. Channel used for this contact. Email, SMS, Push, Social.	3
response_status	varchar(20)	Furthest stage reached for this contact attempt as of last update. Sent, Delivered, Opened, Clicked, Bounced, Unsubscribed, Suppressed.	Opened
suppression_reason	varchar(30)	reason this contact was blocked before sending. NULL if not suppressed. Opt-out, Monthly cap reached.	NULL
was_suppressed	bit	1 if this contact attempt was blocked before sending. 0 if it was sent.	0
sent_date	date	Date the message was sent. NULL if suppressed or not yet sent.	2026-06-01
delivered_date	date	Date the message was confirmed delivered. NULL if not yet known.	2026-06-01
opened_date	date	Date the customer opened the message. NULL if not opened.	2026-06-01
clicked_date	date	Date the customer clicked a link in the message. NULL if not clicked.	NULL

bounced_date	date	Date the message bounced (failed delivery). NULL if not bounced.	NULL
unsubscribed_date	date	Date the customer unsubscribed as a result of this message. NULL if not applicable.	NULL
source_system_code	int	Key of source system. 1 = Marketing Cloud. 2 = Meta. 3 = GA4.	1
create_timestamp	datetime	When the record was created in the data platform.	2026-06-01 0:00:00
update_timestamp	datetime	When the record was last updated. Updates daily as new status data lands from source extract.	2026-06-02 0:00:00

8.2 DIMCAMPAIGN

DimCampaign describes all marketing campaigns across RetailSphere’s three marketing platforms – Salesforce Marketing Cloud, Meta Business Manager, Google Analytics 4. Campaign status is a system-calculated field updated daily by the Silver layer based on end_date comparison. SCD Type 1 is applied – campaign history is not required.

Table 8.2 DimCampaign Dimension Table

Column Name	Data Type	Description	Sample Value
campaign_key	int	Surrogate key of the campaign dimension. Unique, not null, primary key.	200
campaign_id	varchar(20)	Natural key. Format RS-CAM-[SEQUENCE]. Assigned by RetailSphere. Never changes.	RS-CAM-001
campaign_name	varchar(100)	Full name of the marketing campaign.	Winter Warmers 2026
campaign_type	varchar(50)	Type of campaign channel. Email, SMS, Social Media, Display, Push Notification.	Email
budget	money	Total budget allocated to this campaign in ZAR.	R250,000.00
start_date	date	Date the campaign started sending.	2026-06-01
end_date	date	Date the campaign ended. NULL if still active.	2026-06-30
status	varchar(20)	Current campaign status. Planned, Active, Paused, Completed, Cancelled.	Active
source_system_code	int	Key of the source system where this campaign originated. Marketing Cloud, Meta, or GA4.	1
create_timestamp	date	When the record was created in the data platform.	2026-06-01 0:00:00
update_timestamp	date	When the record was last updated in the data platform.	2026-06-01 0:00:00

8.3 DIMCHANNEL

DimChannel is a conformed dimension used on both the Sales and CRM and Marketing domains. Full column design is documented in Section 5.4.

9. FINANCE DOMAIN

The Finance domain tracks all General Ledger transactions for RetailSphere’s financial reporting and audit requirements. FactFinance captures individual GL journal line items enabling Johan’s team to produce month-end close packs, detect backdated postings within 1 hour per FR-FIN-005, and support external auditor requirements. DimAccount introduces the Chart of Accounts hierarchy as a domain-specific dimension.

9.1 FACTFINANCE

FactFinance is a transaction fact table capturing one row per GL journal line item posted in SAP. Two-role playing date dimensions are used – posted_date_key for the date the entry was posted and transaction_date_key for the actual business transaction date. Comparison of these two dates enables automated backdated posting detection and alerting as required by FR-FIN-005.

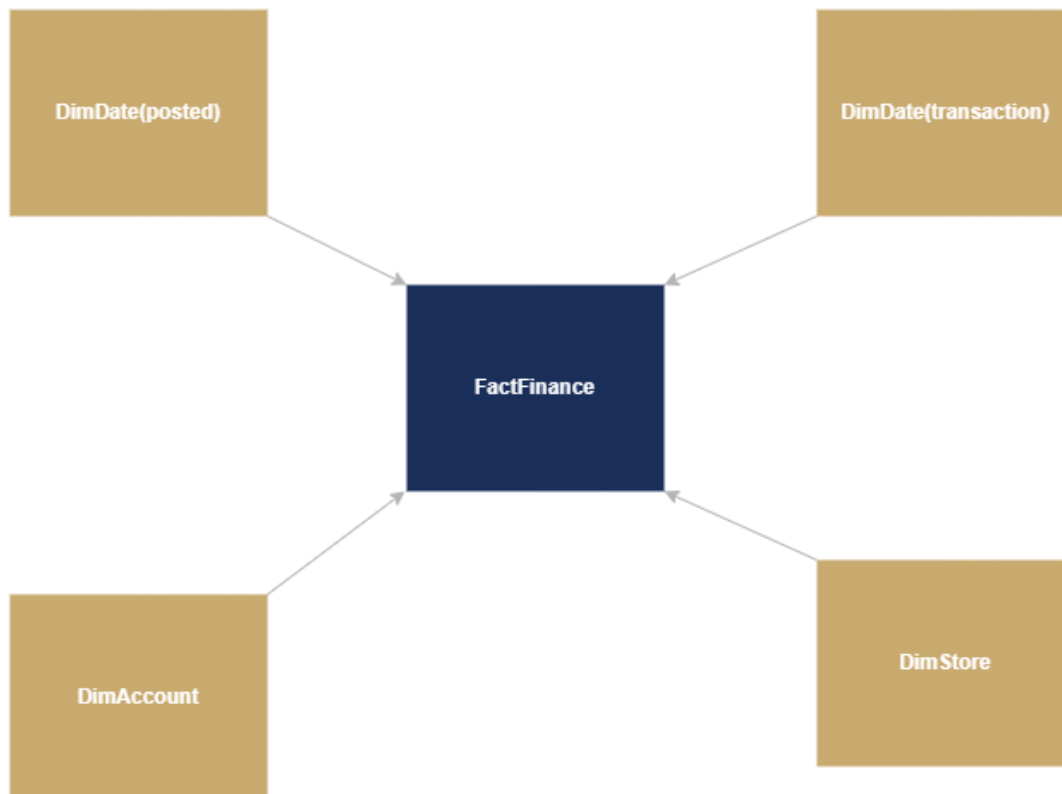


Figure 9.1 – FactFinance Star Schema

Table 9.1 FactFinance Fact Table

Column Name	Data Type	Description	Sample value
posted_date_key	int	FK to DimDate. Date this journal line was posted to the GL in SAP.	20260601
transaction_date_key	int	FK to DimDate. Actual date the underlying business transaction occurred. Compared to posted_date_key to detect backdated postings.	20260528
account_key	int	FK to DimAccount. The GL account this line debits or credits.	300
store_key	int	FK to DimStore. Store this transaction relates to. 0 for head office/non-store transactions.	10
journal_entry_id	varchar(50)	Groups the debit and credit lines belonging to one journal entry. Format RS-JE-[SEQUENCE].	RS-JE-005001
line_number	int	Line number within the journal entry. Starts at 1.	1
debit_amount	money	Amount if this line is a debit. 0 if this line is a credit.	1100
credit_amount	money	Amount if this line is a credit. 0 if this line is a debit.	0

is_backdated	bit	1 if transaction_date falls within a period that was already closed when posted. Triggers FR-FIN-005 alert within 1 hour.	0
source_system_code	int	Key of source system. SAP ECC General Ledger.	1
create_timestamp	datetime	When the record was created in the data platform.	2026-06-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2026-06-01 0:00:00

fiscal_period_key (present in v1.0) has been removed. **fiscal_period** is an attribute of DimDate, not a key column – it is already accessible via the **posted_date_key** join and does not require separate foreign key relationship.

9.2 DIMACCOUNT

DimAccount describes RetailSphere’s Chart of Accounts – the complete list of GL accounts used in financial reporting. It provides a two-level hierarchy of account type and account category enabling financial statement grouping and subtotalling. SCD Type 2 is applied to preserve historical account classifications for audit reproducibility – ensuring that financial statements can be reproduced exactly as originally reported even if account classifications change in future.

Table 9.2 DimAccount Dimension Table

Column Name	Data Type	Description	Sample Value
account_key	int	Surrogate key of the account dimension. Unique, not null, primary key.	300
account_code	varchar(20)	Natural key. SAP General Ledger account number. Never changes once assigned.	4000100
account_name	varchar(100)	Specific name of the GL account as it appears on financial statements.	Clothing Sales Revenue
account_type	varchar(20)	Top-level financial statement classification. Asset, Liability, Equity, Revenue, Expense.	Revenue
account_category	varchar(50)	Grouping within account type. Used for financial statement subtotals.	Sales Revenue
normal_balance	varchar(10)	Whether an increase in this account is normally recorded as a Debit or Credit.	Credit
is_active	bit	1 if account is currently in use. 0 if account has been retired/closed.	1
effective_start_date	date	SCD Type 2 start date. Date from which this version of the account classification is valid. Defaults to 01-01-1900.	2020-01-01
effective_end_date	date	SCD Type 2 end date. NULL means currently active. Defaults to 31-12-2999.	2999-12-31
is_current	bit	1 if this is the current active version of the account record. 0 if expired.	1
source_system_code	int	Key of source system. SAP ECC Chart of Accounts.	1
create_timestamp	datetime	When the record was created in the data platform.	2020-01-01 0:00:00
update_timestamp	datetime	When the record was last updated in the data platform.	2020-01-01 0:00:00

10. Data Mart Summary

This section summarizes the complete dimensional model for the RetailSphere Gold Layer data platform across all six business domains.

Domain	Fact Tables	Dimensions Used
Sales	FactSales	DimDate, DimCustomer, DimProduct, DimStore, DimEmployee, DimChannel
Inventory	FactInventory FactReplenishment	DimDate, DimProduct, DimStore, DimSupplier
Procurement	FactProcurement FactSupplierContract	DimDate, DimProduct, DimSupplier
CRM/Mktg	FactContactHistory	DimDate, DimCustomer, DimCampaign, DimChannel
Finance	FactFinance	DimDate, DimStore, DimAccount

Fact Tables: 7

Dimensions: 9

Total Tables: 16

11. Next Steps

The completion of DM-001 marks the end of Step 7 – Dimensional Model Design. The following steps are required to progress the RetailSphere data platform build:

Step 8 – Source to Target Mapping (STM-001): Map every source system column identified in SSI-001 to its corresponding Gold Layer dimension or fact table column defined in this document.

Step 9 – Physical Database Design (PDD-001): Translate the conceptual dimensional model defined in DM-001 into physical DDL scripts for Microsoft Fabric Warehouse, per ADR-004's medallion architecture decision. Note: Fabric Warehouse constraints are informational only (dbt-enforced); native table partitioning is not currently supported by the platform.

Step 10 – Data Integration Design (DID-001): Design the ADF pipeline architecture for Bronze, Silver, and Gold layer ingestion and transformation, including error handling, retry logic, and monitoring.

Outstanding items requiring resolution before build can begin:

- ADR-002: POPIA legal review and Priya Naidoo sign-off.
- ADR-003: Sipho Dlamini sign-off on PDF extraction approach.
- ADR-006: Rethabile Sithole sign-off on attribution model.
- ADR-007: Priya Naidoo sign-off on contact governance.

12. Document Change Log

Version	Date	Author	Description
1.0	25 June 2026	Thozamile Mbalo	Initial draft. 9 dimension tables and 7 fact tables designed across 6 business domains. DimAccount and FactSupplierContract discovered during design process and added to scope. Star schema diagrams produced in draw.io.
1.1	23 August 2026	Thozamile Mbalo	Corrected FactFinance classification (section 3): accumulating snapshot → transaction fact. Removed fiscal_period_key from FactFinance (section 9.1) — redundant with posted_date_key → DimDate join. Added fact-type exception note for FactSupplierContract (section 7.2): fits none of the three defined types; documented immutability and substitution-handling rule. Corrected stale "Azure Synapse Analytics" partitioning reference in Next Steps, reflecting ADR-004's Fabric Warehouse decision and the platform's current lack of native partitioning support. FactContactHistory attempt-sequencing gap (section 8.1) remains an open item pending validation against Marketing Cloud behaviour — not resolved in this revision.

— End of Document —